6.4.2 Ergodic Stochastic Differential Equations

Now consider an SDE system defined on the space $\mathcal{D} = \mathcal{T}^{k_1} \times \mathbb{R}^{k_2}$ for non-negative integers k_1 and k_2:

$$d\mathbf{X} = \boldsymbol{b}_0(\mathbf{X})dt + \sum_{i=1}^{r} \boldsymbol{b}_i dW_i, \tag{6.43}$$

where $\boldsymbol{b}_0$ is C^∞ and the W_i are independent standard Wiener processes. In this book, we will only consider the analysis for the case of additive noise (i.e. the $\boldsymbol{b}_i$ vectors multiplying a Wiener process are constant). The more complicated case of multiplicative noise (where all of the $\boldsymbol{b}_i$ can vary) will not be needed here, as additive noise is sufficient to sample the canonical distribution.

Working in flat L^2 space, we have that the infinitesmal generator for (6.43) acting on a smooth test function $\phi(\mathbf{X})$ is

$$\mathcal{L}\phi = \boldsymbol{b}_0 \cdot \nabla\phi + \frac{1}{2}\mathrm{Tr}\left(\boldsymbol{B}^T\phi''\boldsymbol{B}\right),$$

where the columns of $\boldsymbol{B}$ are $\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots, \boldsymbol{b}_r$. Taking the adjoint, the forward Kolmogorov or Fokker-Planck operator is $\mathcal{L}^\dagger$, acting on a density $\rho(\mathbf{X})$ as

$$\mathcal{L}^\dagger\rho = -\nabla \cdot (\boldsymbol{b}_0\rho) + \frac{1}{2}\mathrm{Tr}\left(\boldsymbol{B}^T\phi''\boldsymbol{B}\right).$$

In the case of Langevin dynamics, the matrix $\boldsymbol{B}$ is diagonal, simplifying the form of the operators by reducing the term involving second derivatives to a Laplacian.

We have seen that invariant distributions correspond in general to solutions of

$$\mathcal{L}^\dagger\rho_{\mathrm{eq}} \equiv 0.$$

Solutions to this equation are of central importance to the probabilistic behavior of solutions to (6.43). Such solutions give distributions of points in phase space which will remain constant as solutions evolve in time. We may ask a number of questions related to these distributions:

- Is the solution ρ_{eq} unique, up to a constant scaling? (We can see that if ρ_{eq} is a solution, then so is $k \times \rho_{\mathrm{eq}}$ for constant k.)
- If we start with an initial distribution ρ_0 which is different than ρ_{eq}, what sort of behavior do averages with respect to the evolving distribution $\rho(\mathbf{X}, t)$ exhibit? If they converge to a stationary average, at what rate? Does it matter if we start from a distribution "near to" (in a suitable sense) or "far away" from an invariant distribution?